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1 Problems and Requirements of Chemical Protective Clothing

Scenarios where workers are exposed to hazardous materials require chemical protective clothing (CPC). Protection levels range from limited spray protection (Type 6) to fully encapsulating, gas-tight suits (Type 1) with self-contained breathing apparatus (SCBA) ["DGUV Information 212-019"].

Figure 1: Type 1 chemical protection suit 4005/NEO manufactured by isotemp[1]

Confined spaces are defined as areas in which, through their design or site conditions, "inadequate air replacement rates, or the substances [...] found within them or introduced into them, particular hazards exist or may arise that substantially exceed the normal prevailing hazard potential of workplaces" ["DGUV Regel 113-004"]. Tasks that require entering confined spaces are typically cleaning, inspection and/or maintenance of, for example, tanks in chemical plants, rail cars and highway tank trailers and even sewage containers. Traces of the contents may remain, even if their containers are prepared correctly. These residues might—despite being initially inactive—become dangerous through work procedures (e.g.

[1]Vollschutzanzug 4005/NEO · Chemie · isotemp

welding), agitation or chemical reactions [“DGUV Regel 113-004”], hence requiring the highest levels of protection. Other use cases are emergencies, such as confined space rescue, removing chemical spills or other accidents involving unknown, possibly hazardous substances [“OSHA Technical Manual Section VIII”].

These tasks are highly demanding—both physically and mentally—particularly because of the equipment. Since Type 1 suits are not gas-permeable, they compromise the body’s ability to regulate temperature, leading to exhaustion and heat fatigue even faster [Holmér, 2006]. The exertion is exacerbated by heavier equipment and heat stress, ultimately reducing work capacity [Smolander et al., 1984]. Maximum working time in a full-body suit with SCBA is therefore limited and might decrease even more depending on air and exhaustion levels.

Further, the material and shape of the suit muffle hearing and limit mobility and the field of view to such an extent that even trained personnel might need double the time to complete tasks when wearing CPC [Murray et al., 2011].

2 Augmented Reality Technology

Augmented reality (AR) is a subcategory of extended reality (XR), next to mixed reality (MR) and virtual reality (VR). In AR virtual objects are "superimposed upon or composited with the real world" [Azuma, 1997], in opposition to virtual reality which—by replacing the real environment entirely—creates the deepest sense of immersion. Mixed reality combines features of both, the main distinction being the ability to interact with virtual objects in the real world [Hari Chandana et al., 2023].

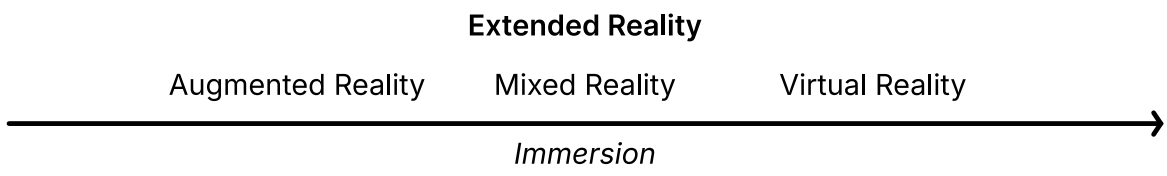

Figure 2: The extended reality spectrum, encompassing augmented, mixed and virtual reality.

Due to not being constrained to two dimensions, XR applications can explore new types of interaction. These technologies share many aspects of interface design and interaction, though each has advantages and disadvantages of its own.

Azuma (1997) defines three requirements for applications to be classified as AR. They combine real and virtual content, they are interactive in real-time, and virtual objects are registered in 3D space. The last criterium was chosen so as to include other technologies besides HMDs. - real and virtual objects "coexist" in the same space, thus requires see-through, contrary to VR

A head-up display overlays information onto reality in the user's line of sight to avoid the need to look away from their primary task to acquire needed information. Originally used in military aircraft, today HUDs are built into consumer vehicles as well. For a more mobile use, the HUD can be implemented as a head-mounted display (HMD).

The displayed information is not anchored to the real world. In the case of a head-worn display, the information is head-stabilized, i.e. 2D planes around the user (see figure 3). The most important feature of head-stabilized content is that it is in the same location every time.

World-stabilized elements are helpful for exploring and better for immersion. When the needed information is unrelated to the environment, head-stabilized display might be better. WHY? Another reason for head-stabilized display is that it is

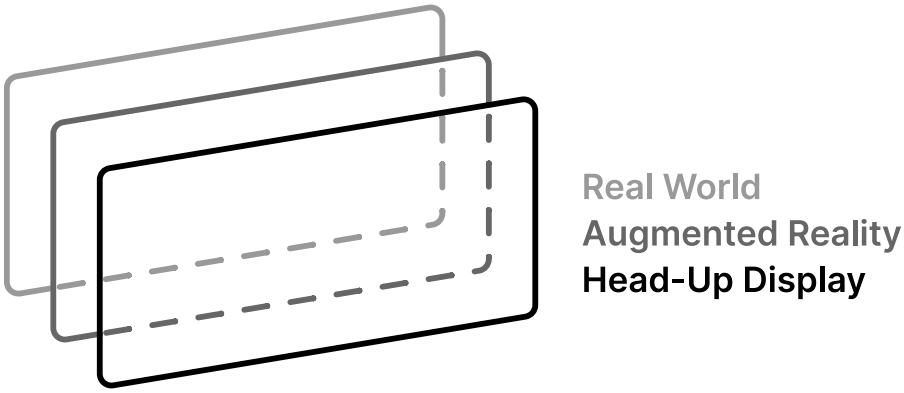

Figure 3: Visualization layers of head-mounted displays: reality, Augmented Reality (3D, world-stabilized) and User Interface (2D, head-stabilized)

hard to orient information when moving around. World-stabilized information might even impact acquisition of information, since the information is not always in the same spatial orientation (e.g. i want info on oxygen, i look to the left).

Manufacturers of AR and VR headsets recommend avoiding classic 2D interfaces, because it breaks immersion. The planned use cases for this HUD require the opposite—to be able to focus on the primary task in the real world.

Apart from built-in HUDs, information can be presented via a head-mounted display. This affixes the display to the head, usually with a helmet or as glasses, enabling mobile usage. The display can cover one (monocular) or both eyes. Depending on the technology, both eyes receive the same image (biocular) or each will be shown a different image (binocular).

HMDs can be monocular, biocular or binocular. In a monocular system, both eyes receive different information, since the HMD is placed in front of only one eye. This introduces the problem of binocular rivalry. With a biocular display, the same image is presented to both eyes, which affects depth perception. Binocular presents a different image to both eyes CAKMAKCI 2006. The depth perception here is affected by the vergence-accomodation conflict. also causes eye strain.

- explain binocular rivalry LARAMEE 2002 - explain how biocular affects depth perception? - vergence-accomodation conflict - ZHOU 2021

2.1 Optical See-Through Displays

There are two options to combine digital content with reality. With video see-through, information will be superimposed on a video stream filmed by a camera on the headset. These headsets are fully occluded, the stream replaces the outside view. An optical see-through system allows the user to directly view reality. Optical combiners

There is no weighing of advantages here. When a video see-through system fails, it will be impossible to see. Failing equipment must not impair vision, especially because it cannot be taken off in a fully encapsulated suit. Other advantages include . - cheaper than VST, better resolution - no eye offset, which hinders depth perception and might cause motion sickness - FOV limitations are cause by distortions of optical tech

Syberfeldt et al. (2017) puts the boundary for an acceptable field of view at 30° at minimum in industrial use cases, which current systems can fulfill. Microsoft's HoloLens sits at 43°x29°, a few systems hit around 40° (https://vr-compare.com/ar). Retinal projection technology does not have limitations on FOV, but is still in development.

As with all head-worn devices, HMDs are suceptible to shaking when walking. Janaka et al. (2023) also found that movement causes legibility issues because of screen shake. Walking can cause the HMD to shake, which impacts legibility. This problem can be avoided by using fonts with thicker strokes (both horizontal and vertical) [Matsuura et al., 2019].

2.1.1 Ambient Light and Contrast Issues

Regardless of optic technology, images are formed by combining light, meaning the display competes with ambient lighting for visibility, which is especially problematic in outdoor conditions. Most established OSTs provide around X lux, compared to overcast winter sky ca. 6000 lux, sunlight > 100.000 lux. The biggest issue for legibility is missing contrast in bright environments. Especially outdoors, visibility of current OST technology is poor. Kerr et al. (2011), for example, report that participants repeatedly used their hands to shield the display from external light to be able to read the content on the display.

This follows the additive color model, where adding all wavelengths of light to-

gether results in white. Black is defined throught the absence of light. With physical screens, the color we see is the actual color of the display material. Since OSTs have no physical screen, "black" is transparent [Gabbard et al., 2013]. This also means that darker colors are unavailable for designing on OSTs, since they are mostly translucent. Many current OSTs like [MODEL] have shaded lenses to improve the contrast somewhat Itoh et al., 2021. Unfortunately, current AR technology is not capable of rendering the contrast levels necessary for outdoor use [Erickson et al., 2020]. Technical solutions in development include OSTs that can partially occlude areas of the display to block light. Still very experimental [Davari et al., 2020].

TANAKA 2008 propose a method to automatically place content where the readability is better, i.e. darker segments of the background. Concepts like these might not be usable in safety-critical use cases, since the user relies on spatially anchored elements for fast acquisition of information.

2.1.2 Color Blending

Because of the display transparency, the background impacts colors. Itoh et al., 2021

GABBARD

2.2 Interaction and Information Access

Eye tracking could be used to dim the HUD while the user does not looks at it. This would decrease distractability, and reduce glare in darker environments. It could also add the capability to summon more information when looking at an anchor. This method was tested by Lu et al. (2020).

Context-Aware interfaces display information based on current task or cognitive load RHODES 1998. The risk of false positives is unjustifiable in safety-critical environments.

2.3 The Concept of Glanceability

Instead of showing the UI at all times, the elements could be positioned outside of the HMD's field of view, to be accessed by either looking or turning the head in the direction. Lu et al. (2020) explored and tested this concept, naming it "Glanceable AR".

- related to gaze-based interaction - head-turning method requires body position tracker to enable sensing the head turning in relation to the body

Would be helpful for secondary information [DEFINE secondary info]. To avoid the user needing to remember in which direction the information was placed, this information could be anchored by representative icons. Or the primary information which hides related information (e.g. task list that shows current and a limited amount of next steps, that displays the complete list when accessed)

LU 2020, 2023 - "designing glanceable interfaces requires careful consideration not only about how to present information visually, but also about how to access the information effectively" Lu et al. (2020) proposes four design principles: "DP1. Virtual content should be fixed to the user. DP2. Virtual content should be spatially distributed in the periphery. DP3. The interaction to access virtual content should be hands-free. DP4. Information should be accessed by glancing quickly at the periphery." - voice input: disturbs others in public, HERE: radio comms, loud industrial env - "we define "periphery" as the space at the edges of or beyond the field of view (FoV) of the AR display" - eye-glance interface (HUD) - baseline, does not require sensors or trackers - "benefits of information being highly visible and accessible, but it may also be obtrusive, occlude the real world, and cause the problem of information overload" - always-there information might interfere with peripheral awareness - head-glance - content outside FOV, fixed to body, turn head to access → clear view - requires to look away from forward FOV momentarily, impacted performance of primary task in study (keeping a specified distance to virtual human when walking) - HYPO might be problematic when moving quickly (emergency) - mental demand higher in study, possibly because of - frequency of information access, leading to a lot of extra head movement avoidable by other methods - need to remember location of information - gaze-summon (look at indicator for 0.5 sec to display information) - additional time needed to access information results in unusability for industrial use case - possible eye strain from keeping gaze on target - increase of false positives when participants took a turn (reflex to look into intended direction + eye-tracking jitter from quick head movements) Lu et al. (2020) mentions the display being less impactful when participants

concentrated on the primary task. CONCENTRATION LEVEL? - CON primary task of walking not comparable to task with high cognitive load + physical demands

Peripheral Displays - there's often alternative forms of information display involved, especially hardware design more resembling art objects than traditional computing devices - ubiquitous computing: computing everywhere in any format

Although peripheral displays are usually mentioned in tandem with ubiquitous computing, relating concepts are applicable here. The first is context awareness of systems, The second, and arguably more relevant, is attention management.

MATTHEWS 2006 - peripheral/ambient displays "move information from the periphery to the center of human attention and back" - "Peripheral displays receive varying amounts of attention depending on how easily users can interpret the information they convey." - this seems interesting - does this mean you pay less attention when the information is easier to parse? - "Glanceability is critical to peripheral display design because users need to quickly glance at and read displayed information with minimal interruption to their primary task." - MATTHEWS 2005 distinguishes between peripheral and notification displays, which are used at the operation and action level respectively. - this concept includes both as "display modes" - "Design variables used for capturing user attention include abrupt onset, flashing, bouncing, and other motion." - "Glanceability is greatly affected by the amount and type of information being displayed." - visual search processes bottom-up (raw sensory input) vs top-down (goal-driven) CONNOR 2004 → that's the vocab for the passive/active modes i had in mind! - "Visuals that support top-down search will be easy to cognitively interpret. Visuals that support bottom-up search enable relevant information to pop-out. Baldassi and Burr, 2004" - "Pop-out is a bottom-up drawing of attention to an object, which occurs when an object within the visual field is distinctive along some visual dimension (for example, possessing a distinctive color or brightness when compared with other objects in the field)." → POWER OF CONTRAST - "If visual features are to be used successfully in glanceable displays, they will need to balance distinctiveness with too much demand for attention." - variables were used to distinguish between versions of otherwise the same items (e.g. email categories of notifications) - task description after Chewar et al. (2004): notification, reaction, comprehension - HYPO icon position for notification dependent on display placement direction (i.e. icons on right side when display is on the left) - information types: ordered, unordered finite, unordered infinite - principles for representation - semantically meaningful - abstract - distinct

Ishiguro and Rekimoto (2011) presented a method to show a simple icon that transitions to related information when gazed at.

Birnholtz et al., 2010 peripheral displays

Notification levels equal importance of information, corresponding to action priority (in HUD case) TABLE notification level and corresponding design choices + elements e.g. change blind, make aware, interrupt, demand action x size, location, color, motion (+ frequency),...

- demand action requires user to perform an action to silence the alert → will not be used in HUD - could argue that oxygen low or hazard proximity demands action, but in in emergency cases there might be no way to immediately take action based on that, and the HUD should not impede the user's decision-making, which an ongoing alarm would do

passive/ambient/top-down By default, the display just shows necessary information. This mode is designed to not get the user's attention so they can focus on their primary task. Another reason to avoid color and movement here is to have the maximum contrast possible if the user needs to get notified of something.

Janaka et al., 2023 getting attention without harshly interrupting primary task Salisbury et al., 2024

interruption/active/notification/bottom-up/ Certain functionality to get the user's attention is implemented. warning pops up in FFOV [Rosenholtz (2016)] + then moves to the side (to not hinder visibility) w/ blinking icon (so it's not forgotten)

3 The Human Visual System and Attention

3.1 The Human Visual System

better sight in low light bc of rods (more populated in peripheral)

A simplified model puts the field of view to around 200° to 220° horizontally and around 125° vertically. The vertical field extends slightly more downwards Loschky et al., 2017.

The area where the angles of both eyes overlap is binocular vision and enables depth perception.

The central vision covers about 5° around the point of fixation, where acuity is highest Loschky et al., 2017, Rosenholtz, 2016.

o foveal/central 1,7 / 5° total or 5° per side?

The remaining field is the periphery, with boundaries at 30° and 60°, resulting in areas of near, mid and far peripheral vision.

(HUD comfort in near peripheral only)

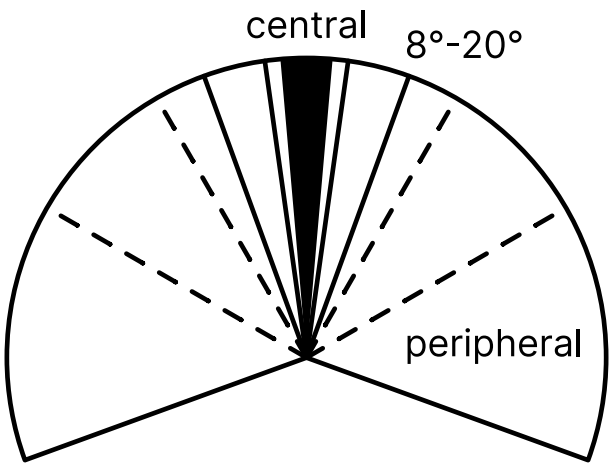

Figure 4: Simplified diagram of the field of view.

Despite changing anatomical structure from center of gaze to periphery, entire scene appears in high resolution, contrast, color saturation Loschky et al., 2017.

Rosenholtz (2016) suggests that crowding is the biggest issue when it comes to parsing information.

GRAPHIC include fig 4 of ROSENHOLTZ 1016? The letters placed close together are perceived as one unit rather than three. Pelli and Tillman, 2008 has a lot of examples

The effect of crowding is also more intense if the target and its neighboring elements are similar (e.g. the same color and shape) Pelli and Tillman, 2008.

Perception does not require a lot of details, the recognition of abjects and orienting oneself ist still possible [Rosenholtz, 2016]. That is why image thumbnails are still recognizable.

Apart from perception, peripheral vision is contributing to other functions, like orientation, navigation, balance and hand-eye coordination. Restricting the peripheral vision affects all of these negatively, getting worse the more is occluded, severely impacting performance Alfano and Michel, 1990. The farther away elements are from the center of vision, the bigger the distance between elements has to be.

Color sensitivity declines toward the periphery, red-green more than blue-yellow and luminance HANSEN 2009. - up to 50° with 8° stimulus size → with established models at 40°, not a problem

3.2 Visual Attention

To cope with the amount of information provided by the eye, filter Rosenholtz, 2016.

The perception of images generates too big an amount of information to process at once. This bottle neck is solved by visual attention: limiting the processing by focusing on just a fragment of the current visual stimuli [Anderson et al., 2005]. by bringing it into the center of vision → saccades

[ ] GRAPHIC perception pipeline: perception, attention as filter, cognition

Rosenholtz, 2016 - "Because of the brain's limited capacity, our visual systems cannot automatically perform every possible visual task at the same time. Instead, the brain may concentrate resources on one task in a given moment, then switch to focus on another task." - "Our results suggest that the visual system looks for evidence of a change throughout the visual field, possibly in parallel, rather than looking for a change merely where one fixates. However, that evidence of a change can be weak because of loss of information in peripheral vision, making change detection difficult."

Saccades: fast, scanning movements Fixations: eye focuses on one spot to comprehend

fixation duration, frequency + how they relate to attention

Chalmers and Cater, 2005 - process by which we humans select a portion of the available visual information for localization, identification, and understanding of objects in an environment - allows the visual system to process visual input preferentially by shifting attention about an image, giving more attention to salient locations and less attention to unimportant regions

HVS ATTENTION - bottom-up: purely stimulus driven, captured automatically without volitional control - top-down: brain directs the eyes to focus on a relevant object

INATTENTIONAL BLINDNESS Research has shown that conspicuous objects in a scene that would normally attract the viewer's attention may be deliberately ignored if they are irrelevant to the visual task at hand

Hou et al., 2025 - "Inattentional blindness occurs when crucial information is disregarded due to the observer's limited attentional resources (Jensen et al., 2011)." - SEC high task perceptual load = more inattention - SEC familiar route = more inattention

Eye movements during perception of complex objects Yarbus A.L. He successfully demonstrated that humans do not scan a scene in a raster-like fashion; rather, the eyes jump to foveate a new point of interest in the scene, called a saccade. What Yarbus also noted was that these saccades were linked to the task or question the viewer had been asked about the scene.

visual system is highly sensitive to features such as edges, abrupt changes in color, and sudden movements → blinking frequency

Interruption

visual perception = reaction Perception describes how brains handle sensory input.

visual attention = recognition attention management = visual attention? Visual attention is defined as a number of continuous eye movements, where the center of vision is the focus on an object.

4 Data Visualization

Visualizating data transforms it into information. Data itself is just noise, whereas information holds value. In what way data needs to be transformed is determined by context [Tergan and Keller, 2005].

The type of visualization depends on the purpose of the information. The main constraint in this case is cognitively demanding situations.

4.1 Situation Awareness and Cognitive Load

As mentioned earlier, the tasks to execute are mentally demanding. Concentrating on fulfilling task objectives while maintaining awareness of the surrounding area result in a high cognitive load, while associated risks prohibit human error. Endsley's model of situation awareness can be referenced to estimate the requirements of information acquisition in dynamic situations.

Endsley (1995) proposes a model of situation awarenes, describing the process of decision-making in dynamic situations. This includes 3 levels, namely perception, comprehension and projection. Only Level 1 awareness—perception—is relevant for UI design itself, though some concepts from Level 2 can be used to exemplify things to avoid.

By displaying relevant information on a HUD, some level of situational awareness could be restored. Freeing up cognitive load, which can then be reallocated. Going by Endsley's definition, that would increase the likelihood of making better decisions.

- "role of goals and goal-directed processing in directing attention and interpreting the significance of perceived information" - circles back to visual attention - goal-directed (top-down) vs data-directed (bottom-up) processing - information salience to grab attention (data-driven) - "people as active participants in the development of their own SA, based on how they direct their attention, communicate with team mates, and manipulate their tools to search for desired information" - "ongoing dynamic process of gathering and interpreting information to update the situation model and using that situation model to search for information until decisions can be made"

- "role of environmental, task, and system factors in SA, including the capabil-

ity of systems for deriving needed information; the effect of system complexity on SA; the **effectiveness of the user interface for providing rapid access and understandability of needed information**; the role of stressors, workload, and fatigue on SA mechanisms and processes; and the effect of automation on SA" ENDSLEY 1995 mentions the advantage of systems providing processed information instead of only the data it is based on (i.e. displaying remaining time based on current oxygen amount and usage). Other recommendations are: - focus types and grouping of information on supporting decision-making for current goal (→ context awareness)

"Cognitive load theory (CLT) is concerned with the design of interfaces that allow users maximize their working memory when problem solving. This includes eliminating any subjective mental load that may be imposed by the instruction interface." DABOR 2019 SIMULT INTERPRETATION

4.2 Types of Information

Depending on the character of the data, different methods of visualization could be chosen.

Comparable to basic dashboard data, only value of the parameter is relevant, since there is no relationship between the data. At most it is the relation between a current and a past value of a parameter.

Purposes of color to cenvey information are grouping or relationships. Grouping is necessary for the HUD, but is managed by the spatial location of and the spacing between elements.

Wilke, 2019 https://learning.oreilly.com/library/view/fundamentals-of-data/9781492031079/ch

4.3 Visualization Methods

5 Design Guidelines

Following scenario requirements, technical limitations and human factors, the HUD needs to fulfill the following demands:

Ideally, the finished system shows as much information as possible/necessary without limiting situation awareness, causing fatigue or sensory overload. The displayed information should be comprehensible in as short a time as possible.

In an emergency, the system must be able to get the user's attention, without hindering decision-making or interrupting the primary task completely.

Referencing rules for visual hierarchy, the usage of contrasts heightens difference between the two modes by limiting color and movement to emergency functions to get and keep attention without hindering reaction times.

The following design guidelines are tailored to ensure the best visibility and information acquisition possible.

Industrial equipment, especially if it might be easily damaged, technology for augmented reality gets more expensive with either bigger field of view or better color rendering. Making sure the proposed design foundations work in monochrome as well enables choosing more economic options, aside from accessibility.

Vicente and Rasmussen, 1990 **novice vs expert users** users are/will be trained with equipment → able to forego certain design principles is there a distinction between end user and professional/trained user? shneiderman (user interfaces): novice + expert user, universal usability HYPO acceptance of HUD is better with newly trained people regarding mental load, since they don't need to change existing patterns

5.1 Color and Contrast

The use of color is mainly limited by current optical see-through technology. As mentioned by GABBARD, colors with high source luminance will be visible best. Tying into economic reasons, designing the HUD functions to work with one color only ensures affordability as well.

Limiting the UI to one color makes sure the design works in all aspects (e.g. using

motion to get attention before color). Another reason is the economic factor: especially in scenarios where it's likely that equipment needs to be replaced because of damage or contamination, the acquisition cost should be as low as possible. One way to achieve this is to only have a monochrome display. So, the HUD UI should be able to function without accent colors. This also avoids impairing users with color blindness.

The library chroma.js can sort colors by luminance in the RGB color space, based on the Web Content Accessibility Guidelines 2.0. Sorting the base colors red (#FF0000), green (#00FF00), blue (#0000FF) and their intermediaries cyan (#00FFFF), magenta (#FF00FF) and yellow (#FFFF00) puts them into the following order (darkest to brightest): blue, red, magenta, green, cyan and yellow.

Figure 5: Base colors sorted by luminance.

https://gka.github.io/chroma.js/#color-luminance

https://www.w3.org/TR/2008/REC-WCAG20-20081211/#relativeluminancedef

https://rapidtoolset.com/en/tool/colors-sorter

These results can be confirmed via the OKLCH 3D model as well: input constant, high lightness and any hue, chroma peaks are yellow, green, cyan. Full chroma is needed to avoid colors blending together.

https://oklch.com/

The following design will use white, since it is the best visible color on an optical see-though display. If color is applicable, yellow will be used as accent color, mainly reserved for notifications, to heighten the contrast with the standard mode. Following Gabbard's study, red seems to be suboptimal, since it is strongly susceptible to color shifting.

5.2 Iconography

Needed information is found quicker when presented as graphical content.

- amount of coded information

Icons can utilize small screen estate more efficently.

- plane-based icons - another reason goes back to MATSUURA's typography study: lines are gonna be harder to read

LIN 2016 "Well-designed graphical icons can achieve such positive effects as drawing users' attention, shortening search time, reducing operational errors, and relieving users' burden." - plane-based icons are recognized easier - line-based icons dissect the space more often = more complex - complexity lowers recognition performance → even more if backgrounds adds another complex layer - white on black icon performed best

JANAKA 2023 - icon-augmented notifcations - significantly higher primary task performance than text - reduced interruption while maintaining reaction and comprehension - when the primary task's complexity increased or needed higher attentional control, users familiar with icons could understand pictogram format better than text format. - In such specialized domains, it is easier to defne a set of well-designed icons that become highly familiar to users through repeated use - icon-augmented notifcations are easier to perceive in suboptimal conditions during mobile OHMD usage. Specifcally, even though virtual content on OHMD is blurred when users focus on the physical world, i.e., limited OHMD focal distance [47, 48], the pictogram format on OHMD can be easier to recognize than the text format - higher salience during shakes, be easier to perceive in sub-optimal conditions (sec 9.2.3), and provide higher attention control

Figure 6: Line-based (left) and plane-based (right) icons.

Especially due to the need of thicker strokes, an icon's readability might be impacted. Simpler icons with varied silhouettes offset this.

Figure 7: Adjustment for the readability of the explosive hazard icon.

5.3 Typography and Stroke Width

Aside from screen shake, reading performance is further impacted by lighting and background texture. To find requirements for font size, RENKEWITZ 2008 tested the effect of "background texture and font color on text recognition". They set a benchmark of two seconds as threshold for fast recognition, which resulted in font size thresholds from 23 to 42 pt, depending on color and background. These numbers might have to be adjusted for an OST-HMD, since they used a video see-through system, and red and blue as colors, both colors with lower luminance.

The typeface "Inter" fulfills all of the requirements. It has a uniform thickness of strokes, square letter width and good spacing between its lines.

Naturally, the same requirements exist for the stroke width.

Figure 8: A comparison of different stroke widths.

5.4 Layout and Spacing

As mentioned in PERIPHERAL VISION, crowding affects information acquisition the most. It seems advisable to prefer wider spacing between elements. limiting the number of elements per side to 3 so they can be spatially anchored to the corners and one slot in between.

Chua et al., 2016 - ! monocular OST-HMD - middle + bottom center positions were noticed quicker - top + peripheral positions were more comfortable, unobtrusive - center-right is mentioned as best balance between usability + comfort

Katsuyama et al., 1989 - studied task requiring frequent eye shifts - (1) displays located 15° below a primary viewing area are better perceived than identical displays located 15° above - (2) ratings of discomfort involved in eye movements (necessitated by the viewing of physically separated displays) correspond to their performances on the secondary task - task performance decreases the further the distance from center

Kim et al., 2019 - tasks that require referencing information - "performance, workload, and usability were more affected by UI designs than the HWD type." - "Compared to the baseline paper pick list condition, using an HWD, especially with the graphic-based UI, led to improved job performance and reduced perceived work-

load, while causing increased discomfort" - "Responses to the workplace cognitive failure scale also indicated a lower risk of cognitive failure specifically with the graphic-based UI design + always-on display."

DYSON 2001

According to Lin et al. (2021), 8° to 20° horizontally seem to be most comfortable. Vertically limit to 8° above and below visual center - "We are surprised to discover that there are no significant differences in task times between the four different display positions. We expected a more uncomfortable viewing angle would have a more dramatic effect in slowing users so that their order picking times would increase, especially for the bottom-right position." BUT "we realized that users stop after picking one book to find and plan their path to pick their next book, instead of trying to read and interpret data while on the move. The extra time taken to stop and plan their path could mask the differences in time taken to perceive picking information" - "the bottom-center position could be inferior in situations in which users have to look at their feet such as stairs mobility and rough terrain maneuvering." - "A bottom-right display could also interfere with tasks involving the hands." - "The display situated between 5.4° to 14.6° toward the ear was most preferred by users. It did not block the line-of-sight view and was also closest to center-center positioning. However, Dowell's work on cognitive capture [8] suggests keeping all pixels at least 8° from line-of-sight." - "there may be an ideal, general purpose HWD position between 8 to 20° temporally offset from line-of-sight. Horizontal FOV would thus be 12°. Vertical FOV requires more investigation, but limiting pixels to between ±8° from center as in Haynes's comfort study [21] seems wise" - "As mentioned previously, from Chua's work we wish to avoid nasally offset displays, and positions much above line-of-sight are uncomfortable for extended use CHUA 2015." - used a monocular display? with 12° x 16° FOV, theorize that they could put another display on the other side, doubling the FOV, and users can switch between → results should be applicable to HUD

Dowell et al., 2002 - flight sim - "HUD information located less than 8 deg from the out-the-window point of interest can induce cognitive tunneling and impair performance on the task over which the HUD is superimposed" - "Additionally, it was shown that HUD symbology placed in the upper portion of the visual scene (greater than 8 deg from the central focus), led to improved simultaneous processing of the HUD symbology and the forward view." BUT "hypothesized that this difference is rooted in the reduced clutter and background motion found when the HUD overlaid the sky, as opposed to the ground."

5.5 Notifications

The billboard style occludes much of reality around the text, which should generally be avoided. However, exactly the feature makes it suitable for high priority notifications.

Matthews et al. (2004) mention three ascpects of attention management in their "Peripheral Displays Toolkit": abstracting information to make it glanceable (see GLANCEABILITY concept), assigning notification levels to information and designing (visual) and transitions between those levels.

Screen edge indicators might not work.

—

Abstraction of Information

optimal information density: shows all relevant information without over whelming the user

GRAPHIC abstraction as scale of readability ↔ unobtrusiveness

Information can be abstracted when the represented concept is simpler (e.g. ambient motion detector), in which case primitive elements of design, such as color and movement can be sufficient.

6 Conceptualization

6.1 Relevant Parameters

A prioritization of elements is necessary for decisions on visual hierarchy, as well as which elements can be left out if there is not enough space.

	Display Mode	Visualization
Oxygen	Continuous	Progress Bar, Value
Hazard Proximity	Pop-Up	Icon
Warning Notification	Pop-Up	Billboard
Working Time	Pop-Up, Continuous	Countdown
Heart Rate	Continuous	Icon, Value, Trend
Ambient Temperature	Continuous	Icon, Value, Trend
Humidity	Continuous	Icon, Value, Trend
Checklist	Continuous	Line Highlight
Tool Help	Pop-Up	Icon, Value

Table 1: Parameters chosen for visualization, ordered from most to least important.

maybe columns: frequency of checking,

Continuous means that the component will be displayed permanently. Parameters marked with pop-up will be displayed occasionally when relevant.

6.2 Components

Following the previous guidelines, every element is designed for maximum clarity.

6.2.1 Visualizing Oxygen Usage Through a Progress Bar

Since the oxygen value is the crucial factor whether or not to continue work, this is the component taking the spotlight. As the most critical parameter, this component will be visible in any version of the HUD. To check the current status as quickly as possible, the value will be visualized with a progress bar.

Visualization helps to avoid mental math.

graphical vs text: progress bar is understood faster than the number amount + unit

The first decision was the way to visualize the progress bar. Mechanical pressure gauges with mechanical pointer show visual progress by default. According to Xia et al. (2025), bar charts outperform radial visualization.

The second version implements the progress bar in four segments, making it easier to estimate the amount left. Another hypothetical advantage of the gaps in the bar might be that more of the background can be seen.

Depending on screen real estate, there are horizontal and vertical versions of the component.

In the horizontal version the label is placed in front of the bar so as not to have another "reading bump".

In the vertical orientation, the label and value are aligned at the top and bottom respectively, to account for visual crowding.

The oxygen value is located so that the progress bar is seen first, e.g. if the component is on the left, the bar will be on the right side. This helps to maintain an optical guide.

- time left w/ current breathing rate

6.2.2 The Base Component

This element is built with a simple configuration for secondary information that is helpful but does not need to be prioritized. This includes factors like heart rate, but also environmental information such as ambient temperature or humidity.

Values that are not to be visualized are similar enough to be based on a universal base component. Apart from the actual parameter value, any of these elements can be hidden. This allows flexible configuration depending on the type of information.

If the icon is unambiguous enough, it is easier to hide the unit. In some cases, the unit supports the identification of the value. Leaving the unit out saves space as well.

The trend arrows are displayed based on the difference of the current and previous value. If the value rises or falls an unusual amount, which could note environmental events. For dynamic elements like these, the frequency of changes should not be too high to avoid distraction.

Comparing the short and long arrows, the long arrows are better readable when out of focus.

Figure 9: The base component with configurable icon, parameter value, unit and trend arrow.

Figure 10: Example configuration of the base component. Left to right: heart rate, time and tool help.

The time component is a special case: it will only be displayed after a configured amount of time. The element popping up is meant to be perceivable in the peripheral vision, marking (for example) the halfway point. This action alone is therefore meaningful. Afterwards it can be checked consciously.

Since working time is limited by various parameters and therefore not fixed, it is a countdown. To avoid pulling unnecessary attention, the display is limited to minutes, instead of ticking down by the second.

6.2.3 Notifications and Pop-Ups

Instead of unique silhouettes, the hazard warnings are triangular to be more intuitive. The placement is therefore more important for peripheral distinction. Both icons are placed close together so that they are visible at the same angle if multiple hazards are present. Similar to a car's dashboard warning lights, each of them has a fixed position, facilitating quicker information acquisition. With regular notifications—depending on which appeared first—the order of content would be different.

Horizontal versus vertical arrangement Since the visual search field is wider than it is tall, horizontal might blend together, stacked vertically—therefore at the same visual angle—they might be more distinctive. More a loose guideline, flexible in regard to screen real estate.

Figure 11: Example notification with billboard style.

The core example of the notification system is the high priority warning. This component is designed with a so-called billboard style, meaning text with a surrounding box. The big, bright area ensures high visibility even if the pop-up is not in the center of vision. Since it is the highest priority, it is the only element placed

in the upper part of the screen. To avoid blocking central vision for too long, the pop-up will transform into a smaller version at the upper edge of the screen.

Grabs attention first, readability second step → text case? Oxygen vs O2/abbreviations better?

6.2.4 The Checklist Component

Since this element is text-based, it relies less on peripheral vision. To anchor the gaze the current task is centered on the screen and more distinctive than the other displayed tasks. Though the other tasks are smaller, everything is left-aligned to—once again—create an optical guide.

sentence case vs lower case

Note that this component is meant as a visual example, which is why certain aspects of functionality might not work in regard to handling the list. (e.g. if only two next tasks are displayed, there is no way to see if new tasks got added → task number display in dynamic situations, generally better for protocol)

The icon to the right is an anchor for possible gaze-based interaction, in this case to show more details. This design keeps the current task centered in the detailed view as well for faster orientation. This also mitigates side effects from possible false positives—i.e. if the system mistakenly opens the detailed view.

6.3 UI Examples

The maximum variant of the interface was designed with full functionality in mind, based on Microsoft's HoloLens 2[2]. This includes a horizontal field of view of 43°, a resolution of 1440 by 936 pixel per eye and full color availability (within the technical limits mentioned in section 2).

Figure 12: Example UI for a binocular HUD.

The layout is based on a three by three grid, anchoring components to the sides and corners, leaving the center free.

[2]

While oxygen is the most important element, it is not necessarily the one that is checked the most, but it should be checkable the quickest. → lower center seems good KATSUYAMA 1989

Occasional information is placed above the oxygen bar, for the same reason: looking down is faster than looking up.

The minimum version is based on the Google Glass, featuring a monocular display with a resolution of 640 by 360 pixels.

https://vr-compare.com/compare?h1=EkSDYv0cW&h2=EnqtMtA05

A point could be made for overlapping components somewhat to use the space better. However, this would disrupt the grid pattern and optical guides, resulting in longer search times.

Figure 13:

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